KAPASA MAKASA UNIVERSITY

FACULTY OF APPLIED SCIENCE AND OPEN LEARNING

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY

ISB 140/DS 110 – INTRODUCTION TO STATISTICS

Tutorial sheet 2

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1. In a competition to grow the tallest holly hock, the height recorded by 50 KMU students from the Department of Information and communication technology were as follows. Heights were measured to the nearest centimetre.

Heights (cm)	**frequency** (f)
$177 - 186$	12
$187 - 191$	8
$192 - 196$	8
$197 - 201$	9
$202 - 206$	7
$207 - 216$	6

   Draw a histogram and superimpose a frequency polygon

2. (a) The first three moments of a distribution about the value 2 of the variable are 1, 16 and – 40.

   (i) Show that the mean is 3, the variance is 15 and $\mu_3 = -86$

   (ii) Also show that the first three moments about $x = 0$ are 3. 24 and 76.

   (b) For a distribution the mean is 10, variance is 16, γ_1 is $+1$ and β_2 is 4. Find the first four moments about the origin.

3. If the standard deviation of a symmetrical distribution is 5. What must be the value of the fourth moment about the mean in order that the distribution be

   (i) leptokurtic, (ii) mesokurtic, and (iii) platykurtic?

4. The numbers 3·2, 5·8, 7·9 and 4·5 have frequencies $x, (x + 2), (x - 3)$ and $(x + 6)$, respectively. If their arithmetic mean is 4·876, find

   (a) the value of *x*.

(b) the Geometric Mean (GM)

(c) the Harmonic Mean (HM)

5. The following table shows the time to the nearest minutes spent reading during a particular day by a group of first year KMU data science students.

Time	Number of students
$10-19$	8
$20-24$	15
$25-29$	25
$30-39$	18
$40-49$	12
$50-64$	7
$65-89$	5

(a) Represent these data by a histogram

(b) Comment on the shape of the distribution.

6. A clothing store bought 8000 shirts last year from five different companies. The following table lists the number of shirts bought from each company and the price paid for each shirt.

Company	Number of Shirts Bought	Price per Shirt (kwacha)
Best shirts	1200	30
Top Wear	1900	45
Smith Shirts	1400	40
New Design	2200	35
Radical Wear	1300	50

Calculate the weighted mean that represents the average price paid for these 8000 shirts.

7. (a) The first three moments of a distribution about the point $X = 7$ are 3, 11 and 15 respectively. Obtain mean, variance and β_1.

(b) The first four moments of distribution about the value 5 of the variable are 2, 20, 40 and 50. Obtain as far as possible, the mean, $\mu_2,\ \mu_3,\ \mu_4,\ \beta_1$ and β_2.

(c) If the first four moments of a distribution about the value 5 are equal to $-4, 22, -117$ and 560, determine the corresponding moment

(i) about the mean and (ii) about zero.

8. (a) For a skewed distribution, the mean is 16, the median is 20 and the standard deviation is 5. Calculate Pearson's coefficient of skewness and sketch the curve.

(b) For a skewed distribution, the mean is 86, the mode is 78 and the variance is 16.

Calculate Pearson's coefficient of skewness and sketch the curve.

9. The following data relate to the percentage unemployment and percentage change in wages over several years.

% unemployment (x)	% change in wages (y)
1.6	5.0
2.2	3.2
2.3	2.7
1.7	2.1
1.6	4.1
2.1	2.7
2.6	2.9
1.7	4.6
1.5	3.5
1.6	4.4

(a) Calculate the product moment correlation coefficient between x and y.

(b) It has been suggested that low unemployment and a low rate of wage inflation cannot exist together. Without further calculation, use your correlation coefficient to explain briefly whether or not you think the suggestion is justified.

10. Calculate the weighted arithmetic average from the following data and comment on them:

Designation	Monthly salary (in kwacha)	Strength of the cadre
Class I officers	1500	10
Class II officers	800	20
Subordinate staff	500	70
Clerical staff	250	100
Lower staff	100	150

11. (a) The first four moments of a distribution about $x = 4$ are 1, 4, 10, 45. Show that the mean is 5 and the variance is 3 and μ_3 and μ_4 are 0 and 26 respectively,

(b) In certain distribution, the first four moments about the point 4 are $-1 \cdot 5, 17, -13$ and 308. Calculate β_1 and β_2

12. The following grouped frequency distribution summarises the time to the nearest minute spent waiting by a sample of patients in a doctor's surgery. given that the mean of the times was 9.63 minutes and the standard deviations was 3.03 minutes.

Waiting Time	Number of Patients
3 or less	6
$4-6$	15
$7-8$	27
9	49
10	52
$11-12$	29
$13-15$	13
16 or more	9

(a) calculate the coefficient of variation of the data

(b) Estimate the median and the semi-interquartile range.

13. Find out the kurtosis of the data given below:

Class Intervals	$0-10$	$10-20$	$20-30$	$30-40$
Frequency	1	3	4	2

14. Ten female students of KMU complete in throwing a cricket ball, and the table below shows the height of each female student (x cm) measured to the nearest centimetre and the distance (y cm) to which she can throw the ball.

Student	(X)	Y
A	122	41
B	124	38
C	133	52
D	138	56
E	144	29
F	156	54
G	158	59
H	161	61
I	164	63
J	168	67

Calculate the product moment correlation coefficient. Comment on the result.

15. The following table gives the blood pressure of 60 ISB students at Kapasa Makasa University (KMU).

Blood pressure	frequency
$95 -$	2
$105 -$	5
$110 -$	6
$115 -$	9
$120 -$	14
$125 -$	3
$130 -$	6
$135 -$	5
$140 -$	4
$150 - 180$	6

(a) Find

(i) Pearson's coefficient of skewness

(ii) The quartile coefficient of skewness

(b) Draw the histogram and comment on the shape of distribution.

16. For the following data, calculate (i) Semi-inter-quartile range, (ii) Coefficient of variation, and (iii) β_1 and β_2 coefficients.

Wages in kwacha	$170 - 180$	$180 - 190$	$190 - 200$	$200 - 210$	$210 - 220$	$220 - 230$	$230 - 240$	$240 - 250$
No. of persons	52	68	85	92	100	95	70	28

17. (a) The following data give the total food expenditures (in kwacha) for the past one month for a sample of 20 families.

1125 530 1234 595 427 872 1480 699 1274 1187

933 1127 716 1065 934 930 1046 1199 1353 441

Calculate the 20% trimmed mean for these data.

(b) The following data give the annual salaries (in thousand dollars) of 20 randomly selected health care workers.

50 71 57 39 45 64 38 53 35 62

74 40 67 44 77 61 58 55 64 59

Calculate the 15% trimmed mean for these data.

18. Find Pearson's coefficient of skewness for the distribution represented by the stem and leaf plot which gives marks in the examination.

Stem	Leaf
1	9
2	2 8
3	3 7
4	5
5	2 5 5 7
6	1 1 6 6 8 8 8
7	3 5 5
8	2 9
9	1

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19. (a) Suppose a sample of 10 statistics books gave a mean price of k140 and a sample of 8 mathematics books gave a mean price of k160. Find the combined mean.

(b) The mean 2015 income for five families was k99,520. What was the total 2015 income of these five families?

(c) The mean age of six persons is 46 years. The ages of five of these six persons are 57, 39, 44, 51, and 37 years, respectively. Find the age of the sixth person.

20. Over a period of four years a bank keeps a weekly record of the number of cheques with errors that are presented for payments. The results for the200 accounting weeks are as follows.

Number of cheques with errors (x)	Number of weeks (f)
0	5
1	22
2	46
3	38
4	31
5	23
6	16
7	11
8	6
9	2

(a) Construct a suitable pictorial representation of these data

(b) State the modal value and calculate the median, mean and standard deviation of the number of cheques with errors in a week.

(c) Some text books measure the skewness (or asymmetry) of a distribution by

$$\frac{3(mean - median)}{standard\ deviation}$$

And others measure it by

$$\frac{(mean - mode)}{standard\ deviation}$$

Calculate and compare the values of these two measures of skewness for the above data.

(d) State how this skewness is reflected in the shape of your graph.

21. (a) The first three moments of distribution about the value 2 are 1, 16 and 40 respectively. Examine the skewness of the distribution.

(b) A frequency distribution gives the following results: $C.V. = 5$, Karl Pearsons's coefficient of skewness $= 0 \cdot 5$ and $\sigma = 2$. Find the mean and mode of the distribution.

(c) Find the $C.V$ of a frequency distribution given that its mean is 120, mode is 123 and Karl Pearsons's coefficient of skewness is $-0 \cdot 3$.

22. Twelve students were given a prognostic test at the start of the course and their score X in the test were compared with their scores Y obtained in an examination at the end of the course.

Student	(X)	Y
A	1	3
B	2	4
C	2	5
D	4	5
E	5	4
F	5	8
G	6	6
H	7	6
I	8	6
J	8	7
K	9	8
L	9	10

Determine the product moment correlation coefficient and briefly comment on your answer.

23. Find the second, third and fourth central moments of the frequency distribution given below. Hence find (i) a measure of skewness γ_1) and (ii) measure of kurtosis (γ_2).

Class Limits	Frequency
$100.0 - 114.9$	5
$115.0 - 119.9$	15
$120.0 - 124.9$	20
$125.0 - 129.9$	35
$130.0 - 134.9$	10
$135.0 - 139.9$	10
$140.0 - 144.9$	5